

SOHAM RAVINDRA LOKHANDE

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Education

University of California, San Diego - Master of Science in Computer Science

Expected June 2028

University of California, Irvine - Bachelor of Science in Computer Science - GPA: 3.94 (Cum Laude)

2021 - 2025

Relevant Coursework: Artificial Intelligence, Machine Learning and Data Mining, Information Retrieval, Operating Systems, Computer Networks, Distributed Systems, Relational Database Management, IoT Software & Systems, Data Structures & Algorithms.

Experience

Machine Learning Researcher - Algovorse (San Francisco, CA)

May 2026 - Present

- Led a 3-person team to build an ML quality-control system for CT/MRI organ segmentations, reducing manual review workload by 25%; research accepted to the **NeurIPS 2026 GenAI4Health Workshop Conference**.
- Architected an **ML pipeline in Python using scikit-learn** with 5 idempotent, resumable components (ingestion, feature engineering, curation, training, evaluation), engineering 19 geometric/intensity mask features with organ-relative normalization.
- Validated model robustness through feature-ablation studies on a **SLURM-managed HPC cluster**, maintaining 90% accuracy and confirming the classifier learns genuine mask-quality signal.

Software Engineer - TakeBridge (San Francisco, CA)

May 2025 - Feb 2026

- Led 0 to 1 build of an Agentic automation platform combining LLM agents + human-in-the-loop operator, delivering in **Agile sprints**.
- Owned 2 core async **Python FastAPI REST** services for workflow execution and real-time updates, supporting 1,000+ concurrent requests.
- Built a control-plane service to provision and manage user-scoped **AWS EC2 sandboxes**, enabling isolated execution environments and improving platform reliability by 30%.
- Designed the persistence and artifact layer with **PostgreSQL** for workflow state and **Cloudflare R2** for artifacts, reducing stored state size by 50% and enabling recoverable runs.

Software Engineer Intern - Cognitus, an IBM Company (Dallas, TX)

June 2024 - Sep 2024

- Architected and shipped features for a **Python FastAPI microservice** within an AI-powered CLM platform used by **10+ SAP federal contracting clients**, driving delivery in **Agile sprints** with **GitLab PR** reviews.
- Engineered and validated 30+ mission-critical REST APIs backed by **PostgreSQL + Redis**, and integrated **Docker Compose** testing into **GitLab CI/CD**, reducing workflow/UI inconsistencies by 25%.
- Designed a multilingual (i18n) API messaging framework via middleware + message-key standardization, enabling localized responses across 30+ endpoints for 5+ international clients.

Projects

TakeBridge - Human-in-the-Loop Agentic Automation Platform [GitHub](#)

Python | LangChain | LLM | AWS EC2 | Docker

- Built a B2B SaaS that automates repetitive back-office business workflows across multiple apps, saving 6+ hours of manual effort per workflow run.
- Orchestrated LLM agents that combine **MCP tool adapters** with computer-use (GUI) actions, supported by a human-in-the-loop operator console and real-time execution streaming.
- Ran workflows in isolated **AWS EC2 sandboxes** using **PostgreSQL** for state, **Cloudflare R2** for artifacts, and **vLLM** for self-hosted inference.

ASRA - Market Research Automation GenAI Agent [GitHub](#)

Python | Next.js | LangGraph | ChromaDB | Docker

- Built a multi-agent GenAI research system that turns startup ideas into evidence-backed market reports from **100+ sources/run**.
- Implemented a **RAG pipeline** using OpenAI embeddings and **ChromaDB**, with an **80% relevance gate** to filter noise and improve evidence quality.
- Added memory/context controls to **reduce token usage by 50%** while preserving research coverage.

GenResume - Job-Specific Resume Recommendation Engine [GitHub](#)

Python | PostgreSQL | React | AWS | RAG

- Built a **GitHub OAuth** web app that ranks repositories into job-specific bullets, cutting resume tailoring time by 80%.
- Designed a Tree-sitter and pgvector semantic ranking pipeline, achieving 85% accuracy in aligning evidence with job requirements.
- Deployed to a **Kubernetes cluster on AWS** with a production backend, **PostgreSQL database** and **React UI**.

Technical Skills

Languages: Python, Java, C++, C, JavaScript, SQL, R, HTML, CSS, TypeScript

Developer Tools: Git, AWS, Postman, Docker, Kubernetes, Azure, Jira, GCP, Linux, vLLM

Libraries/Frameworks: FastAPI, Flask, React, Next.js, SQLAlchemy, Tailwind CSS, NumPy, Pandas, TensorFlow, PyTorch, Redis, Node.js, JWT, OAuth 2.0, LangGraph, Matplotlib